

1. (Fall 2024, #2) Compute the indefinite integrals.

$$(a) \int (x^2 - x + 1)e^x dx \quad \int u dv = uv - \int v du$$

$$\begin{aligned} (a) &= \int (x^2 - x + 1) de^x \\ &= (x^2 - x + 1)e^x - \int (2x - 1)e^x dx \\ &= (x^2 - x + 1)e^x - \int (2x - 1) de^x \\ &= (x^2 - x + 1)e^x - \left((2x - 1)e^x - \int 2e^x dx \right) \\ &= (x^2 - x + 1)e^x - (2x - 1)e^x + 2e^x + C \\ &= e^x (x^2 - 3x + 4) + C. \end{aligned}$$

$$(b) \int \sin^3 x dx$$

$$(b) = \int -\sin^2 x \cdot (-\sin x) dx$$

$$= \int -\sin^2 x d\cos x$$

$$= \int (\cos^2 x - 1) d\cos x$$

$$\stackrel{u=\cos x}{=} \int (u^2 - 1) du$$

$$= \frac{1}{3} u^3 - u + C.$$

$$= \frac{1}{3} \cos^3 x - \cos x + C.$$

2. (Fall 2023, #2) Compute the following antiderivatives.

$$(a) \int x(x^2+1) \ln(x^2+1) dx$$

$$(a) = \frac{1}{2} \int (x^2+1) \ln(x^2+1) d(x^2+1)$$

$$\stackrel{u=x^2+1}{=} \frac{1}{2} \int u \ln u du$$

$$= \frac{1}{2} \int \ln u d \frac{u^2}{2}$$

$$= \frac{1}{2} \ln u \cdot \frac{u^2}{2} - \frac{1}{2} \int \frac{1}{u} \cdot \frac{u^2}{2} du.$$

$$= \frac{1}{4} \ln u \cdot u^2 - \frac{1}{2} \int \frac{u}{2} du.$$

$$= \frac{1}{4} \ln u \cdot u^2 - \frac{1}{4} \cdot \frac{u^2}{2} + C.$$

$$= \frac{1}{4} \ln(x^2+1) (x^2+1)^2 - \frac{1}{8} (x^2+1)^2 + C.$$

$$(b) \int \frac{\sqrt{t^2-9}}{t^6} dt$$

$$\stackrel{t=3\sec\theta}{=} \int \frac{\sqrt{9\sec^2\theta-9}}{(3\sec\theta)^6} 3\sec\theta \tan\theta d\theta.$$

$$= \int \frac{3\tan\theta}{3^6 (\sec\theta)^6} \sec\theta \tan\theta d\theta.$$

$$= \int \frac{1}{3^5} \cdot \frac{\tan^2\theta}{(\sec\theta)^5} d\theta.$$

Recall:

① Trig-substitution

$$\sqrt{t^2-a^2} \quad \text{let } t = a\sec\theta.$$

$$\textcircled{2} (\sec\theta)' = \sec\theta \tan\theta$$

$$\textcircled{3} 1 + \tan^2\theta = \sec^2\theta.$$

$$= \int \frac{1}{3^5} \cos^5 \theta \cdot \frac{\sin^2 \theta}{\cos^2 \theta} d\theta .$$

$$= \int \frac{1}{3^5} \cdot \sin^2 \theta \cdot \cos^3 \theta d\theta .$$

$$= \int \frac{1}{3^5} \sin^2 \theta \cos^2 \theta d\sin \theta .$$

$$\stackrel{u=\sin \theta}{=} \int \frac{1}{3^5} u^2 (1-u^2) du .$$

$$= \frac{1}{3^5} \int u^2 - u^4 du .$$

$$= \frac{1}{3^5} \left(\frac{1}{3} u^3 - \frac{1}{5} u^5 \right) + C .$$

$$= \frac{1}{3^5} \left(\frac{1}{3} (\sin \theta)^3 - \frac{1}{5} (\sin \theta)^5 \right) + C .$$

Last step: $t = 3 \sec \theta \Rightarrow \cos \theta = \frac{3}{t} \Rightarrow \sin \theta = \sqrt{1 - \frac{9}{t^2}} .$

$$= \frac{1}{3^5} \left(\frac{1}{3} \left(\sqrt{1 - \frac{9}{t^2}} \right)^3 - \frac{1}{5} \left(\sqrt{1 - \frac{9}{t^2}} \right)^5 \right) + C .$$

3. (Spring 2023, #4) Evaluate $\int \frac{x^2}{(\sqrt{1-4x^2})^3} dx$.

$$\underline{x = \frac{1}{2} \sin \theta} \quad \int \frac{\frac{1}{4} \sin^2 \theta}{(\cos \theta)^3} \cdot \frac{1}{2} \cos \theta d\theta .$$

$$= \frac{1}{8} \int \frac{\sin^2 \theta}{\cos^2 \theta} d\theta .$$

$$= \frac{1}{8} \int \frac{1 - \cos^2 \theta}{\cos^2 \theta} d\theta$$

$$= \frac{1}{8} \int (\sec^2 \theta - 1) d\theta .$$

Recall:

$$(\tan \theta)' = \sec^2 \theta \quad = \frac{1}{8} \tan \theta - \frac{1}{8} \theta + C .$$

$$= \frac{1}{8} \frac{2x}{\sqrt{1-4x^2}} - \frac{1}{8} \arcsin(2x) + C .$$

$$\sin \theta = 2x$$

$$\cos \theta = \sqrt{1-4x^2}$$

$$\tan \theta = \frac{2x}{\sqrt{1-4x^2}}$$

$$= \frac{x}{4\sqrt{1-4x^2}} - \frac{1}{8} \arcsin(2x) + C .$$

4. (Fall 2022, #2) Compute the following indefinite and definite integrals:

(a) $\int x(\ln x)^2 dx$

$$= \int (\ln x)^2 d \frac{x^2}{2}$$

$$= (\ln x)^2 \cdot \frac{x^2}{2} - \int 2 \ln x \cdot \frac{1}{x} \cdot \frac{x^2}{2} dx.$$

$$= (\ln x)^2 \cdot \frac{x^2}{2} - \int x \ln x dx$$

$$= (\ln x)^2 \frac{x^2}{2} - \int \ln x d \frac{x^2}{2}$$

$$= (\ln x)^2 \frac{x^2}{2} - \left(\ln x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx \right).$$

$$= (\ln x)^2 \frac{x^2}{2} - \frac{\ln x}{2} \cdot x^2 + \int \frac{x}{2} dx = (\ln x)^2 \frac{x^2}{2} - \frac{\ln x}{2} \cdot x^2 + \frac{x^2}{4} + C.$$

(b) $\int_{-1}^1 (1-x)\sqrt{1-x^2} dx$

$$= \int_{-1}^1 \sqrt{1-x^2} dx - \int_{-1}^1 x \sqrt{1-x^2} dx$$

0 without computation.

Why? (odd function).

$$= \int_{-1}^1 \sqrt{1-x^2} dx$$

$$\stackrel{x=\sin \theta}{=} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta \cos \theta d\theta.$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 \theta d\theta.$$

$$\stackrel{\text{half-angle}}{=} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos 2\theta + 1}{2} d\theta = \frac{\sin 2\theta}{4} + \frac{\theta}{2} \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{\pi}{2}.$$

5. (Fall 2020, #2) Evaluate the following integrals.

(a) $\int e^{1-ax} \sin(1+ax) dx$ where a is any real number.

$$= \frac{1}{-a} \int \sin(1+ax) d e^{1-ax}$$

$$= \frac{1}{-a} \sin(1+ax) e^{1-ax} - \frac{1}{-a} \int a \cdot \cos(1+ax) e^{1-ax} dx$$

$$= -\frac{1}{a} \sin(1+ax) e^{1-ax} + \int \cos(1+ax) e^{1-ax} dx -$$

$$= -\frac{1}{a} \sin(1+ax) e^{1-ax} - \frac{1}{a} \int \cos(1+ax) d e^{1-ax}$$

$$= -\frac{1}{a} \sin(1+ax) e^{1-ax} - \frac{1}{a} \left(\cos(1+ax) e^{1-ax} - \int a \sin(1+ax) e^{1-ax} dx \right)$$

$$= -\frac{1}{a} \sin(1+ax) e^{1-ax} - \frac{1}{a} \cos(1+ax) e^{1-ax} - \boxed{\int \sin(1+ax) e^{1-ax} dx}$$

$$\Rightarrow \text{Ans} = \frac{1}{2} \left(-\frac{1}{a} \sin(1+ax) e^{1-ax} - \frac{1}{a} \cos(1+ax) e^{1-ax} \right) + C$$

(b) $\int \frac{x}{\sqrt{4x^2 + 4x + 5}} dx$

$$= \int \frac{x}{\sqrt{(2x+1)^2 + 4}} dx$$

$$\text{Let } 2x+1=u \Rightarrow 2dx=du \Rightarrow dx=\frac{1}{2} du$$

$$= \int \frac{\frac{u-1}{2}}{\sqrt{4+u^2}} \cdot \frac{1}{2} du$$

$$= \frac{1}{4} \int \frac{u-1}{\sqrt{4+u^2}} du$$

$$\stackrel{u=2\tan\theta}{=} \frac{1}{4} \int \frac{2\tan\theta-1}{2\sec\theta} \cdot 2\sec^2\theta d\theta$$

$$= \frac{1}{4} \int (2\tan\theta-1)\sec\theta d\theta$$

$$= \frac{1}{2} \int \tan\theta \sec\theta d\theta - \frac{1}{4} \int \sec\theta d\theta$$

$$= \frac{1}{2} \int \frac{\sin\theta}{\cos^2\theta} d\theta - \frac{1}{4} \ln|\sec\theta + \tan\theta| + C$$

$$= \frac{1}{2} \int \frac{-1}{\cos^2 \theta} d\cos \theta - \frac{1}{4} \ln |\sec \theta + \tan \theta| + C$$

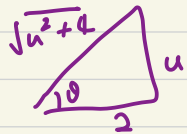
$$v = \cos \theta$$

$$= \frac{1}{2} \int \frac{-1}{v^2} dv - \frac{1}{4} \ln |\sec \theta + \tan \theta| + C$$

$$= \frac{1}{2} \frac{1}{v} - \frac{1}{4} \ln |\sec \theta + \tan \theta| + C$$

$$= \frac{1}{2} \sec \theta - \frac{1}{4} \ln |\sec \theta + \tan \theta| + C$$

$$\tan \theta = \frac{1}{2} u$$



$$\sec \theta = \frac{\sqrt{u^2 + 4}}{2}$$

$$= \frac{1}{2} \cdot \frac{\sqrt{u^2 + 4}}{2} - \frac{1}{4} \ln \left| \frac{\sqrt{u^2 + 4}}{2} + \frac{u}{2} \right| + C$$

$$u = 2x+1$$

$$= \frac{1}{2} \frac{\sqrt{(2x+1)^2 + 4}}{2} - \frac{1}{4} \ln \left| \frac{\sqrt{(2x+1)^2 + 4}}{2} + \frac{2x+1}{2} \right| + C$$